

Grade 12 Trig Cycle Test 29 April 2024

Question 1

1.1. $\sin 2\beta = \frac{3}{-5}$

$$x^2 + y^2 = r^2$$

$$x^2 + (-3)^2 = (5)^2$$

$$x^2 = 16$$

$$x = -4, \text{ ✓ value of } x$$

$$\cos 2\beta = -\frac{4}{5} \text{ ✓ CA}$$

(4)

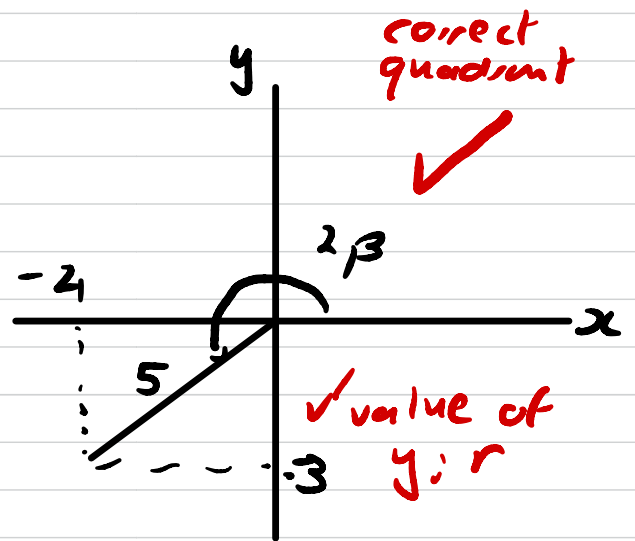
1.2 $2 \cos^2 \beta - 1 = -\frac{4}{5} \text{ ✓ M}$

$$2 \cos^2 \beta = \frac{1}{5}$$

$$\cos^2 \beta = \frac{1}{10} \text{ ✓ CA}$$

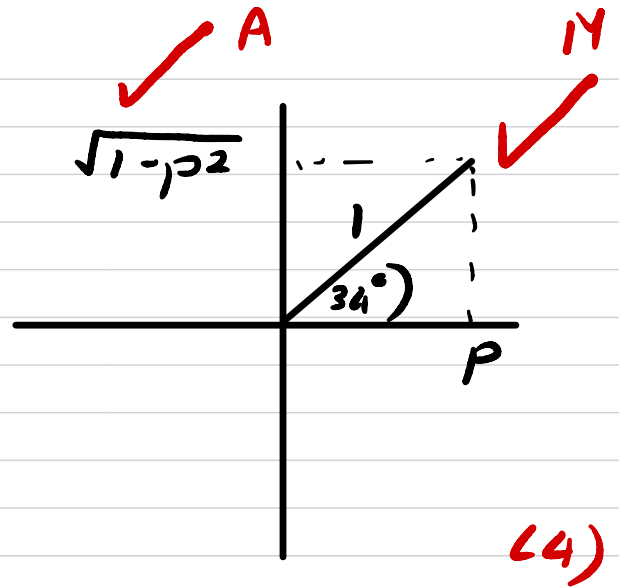
$$\cos \beta = -\sqrt{\frac{1}{10}} \text{ ✓ CA}$$

(3)



Question 2

$$\begin{aligned} 2.1. \quad & \tan 214^\circ \\ &= \tan (180^\circ + 34^\circ) \\ &= \tan 34^\circ \quad \checkmark A \\ &= \frac{\sqrt{1 - p^2}}{p} \quad \checkmark CA \end{aligned}$$



$$\begin{aligned} 2.2. \quad & \sin 68^\circ \\ &= 2 \sin 34^\circ \cos 34^\circ \quad \checkmark M \\ &= 2 \sqrt{1 - p^2} \cdot p \quad \checkmark CA \quad \checkmark A \end{aligned}$$

(3)

$$\begin{aligned} 2.3. \quad & \sin 26^\circ \\ &= \sin (60^\circ - 34^\circ) \quad \checkmark M \\ &= \sin 60^\circ \cos 34^\circ - \cos 60^\circ \sin 34^\circ \quad \checkmark A \\ &= \frac{\sqrt{3}}{2} \cos 34^\circ - \frac{1}{2} \sin 34^\circ \quad \checkmark A \text{ special angle values} \\ &= \frac{\sqrt{3}}{2} p - \frac{1}{2} \sqrt{1 - p^2} \quad \checkmark CA \end{aligned}$$

(4)

Question 3

$$3.1. \sin(90^\circ - x) \cdot \cos(-x) + \cos(90^\circ + x) \sin(360^\circ + x)$$

$\checkmark A$ $\checkmark A$ $\checkmark A$ $\checkmark A$

$$= \cos x \cdot \cos x - \sin x \sin x$$

$$= \cos^2 x - \sin^2 x$$

$$= \cos 2x \quad \checkmark CA$$

15)

$$3.2. \quad \frac{\cos 27^\circ}{\sin 21^\circ} - \frac{\sin 27^\circ}{\cos 21^\circ}$$

$$= \frac{\cos 27^\circ \cos 21^\circ - \sin 27^\circ \sin 21^\circ}{\sin 21^\circ \cos 21^\circ} \quad \checkmark LCD$$

$$= \frac{\cos(27^\circ + 21^\circ)}{\frac{1}{2} \cdot 2 \sin 21^\circ \cos 21^\circ} \quad \checkmark \text{compound angle}$$

$\checkmark M$

$$= \frac{\cos 48^\circ}{\frac{1}{2} \sin 42^\circ} \quad \checkmark \text{double angle}$$

$$= \frac{2 \cos 48^\circ}{\cos 48^\circ} \quad \checkmark \text{co function}$$

$$= 2$$

5

Question 4

$$\begin{aligned} 4.1. \quad \text{LHS:} & \quad \frac{1}{\cos 2x} + \tan 2x \\ &= \frac{1}{\cos 2x} + \frac{\sin 2x}{\cos 2x} \checkmark \\ &= \frac{1 + \sin 2x}{\cos 2x} \checkmark \\ &= \frac{1 + 2\sin x \cos x}{\cos^2 x - \sin^2 x} \checkmark \\ &= \frac{\cos^2 x + \sin^2 x + 2\sin x \cos x}{\cos^2 x - \sin^2 x} \checkmark \\ &= \frac{\cos^2 x + 2\sin x \cos x + \sin^2 x}{\cos^2 x - \sin^2 x} \\ &= \frac{(\cos x + \sin x)^2}{(\cos x - \sin x)(\cos x + \sin x)} \checkmark \\ &= \frac{\cos x + \sin x}{\cos x - \sin x} \checkmark \end{aligned}$$

(6)

$$4.2. \quad \cos 2x = 0^\circ$$

$$2x = 90^\circ + k \cdot 180^\circ ; k \in \mathbb{Z}$$

$$x = 45^\circ + k \cdot 90^\circ$$

$\tan 2x$ undefined

$$2x = 90^\circ + k \cdot 180^\circ ; k \in \mathbb{Z}$$

$$x = 45^\circ + k \cdot 90^\circ$$

$$\cos x - \sin x = 0$$

$$\sin x = \cos x$$

$$\tan x = 1$$

$$x = 45^\circ + k \cdot 180^\circ ; k \in \mathbb{Z}$$

$$x \in \left\{ -135^\circ ; -45^\circ ; 45^\circ ; 135^\circ \right\} \quad (4)$$

Question 5

$$5.1. \quad \cos 47^\circ - \cos 13^\circ$$

$$= \cos(30^\circ + 17^\circ) - \cos(30^\circ - 17^\circ) \quad \checkmark M$$

$$= \cos 30^\circ \cos 17^\circ - \sin 30^\circ \sin 17^\circ - (\cos 30^\circ \cos 17^\circ + \sin 30^\circ \sin 17^\circ) \quad \checkmark A$$

$$= \frac{\sqrt{3}}{2} \cos 17^\circ - \frac{1}{2} \sin 17^\circ - \frac{\sqrt{3}}{2} \cos 17^\circ - \frac{1}{2} \sin 17^\circ \quad \checkmark \text{ special angles}$$

$$= -\sin 17^\circ \quad \checkmark CA$$

(5)

$$5.2. \quad \cos x = \cos 47^\circ - \cos 13^\circ$$

$$\cos x = -\sin 17^\circ$$

$$\cos x = -\cos 73^\circ \quad \checkmark M$$

$$x = 180^\circ - 73^\circ + k \cdot 360^\circ ; k \in \mathbb{Z}$$

$$x = 107^\circ + k \cdot 360^\circ \quad \checkmark CA$$

OR

$$x = 180^\circ + 73^\circ + k \cdot 360^\circ ; k \in \mathbb{Z}$$

$$x = 253^\circ + k \cdot 360^\circ \quad \checkmark CA$$

(3)

Question 6

$$(\sin x + \cos x)^2 = \left(\frac{2}{3}\right)^2 \quad \checkmark M$$

$$\sin^2 x + 2\sin x \cos x + \cos^2 x = \frac{4}{9} \quad \checkmark M$$

$$\sin^2 x + \cos^2 x + \sin 2x = \frac{4}{9}$$

$$1 + \sin 2x = \frac{4}{9} \quad \checkmark A$$

$$\sin 2x = -\frac{5}{9} \quad \checkmark CA \quad (4)$$